

Lesson 4-3 · Period 3

Rational Expression Modeling + Assessment-Critical Simplify

Klimesara-adapted · Student-centered · No Blooket

DOK 2

5 min

bridge from P2

Do Now · Identity check (4-3-savvas-q11) Reason Explain why

$$\frac{4x^2 - 7}{4x^2 - 7} = 1$$

is a valid identity — and state the values of x that must be restricted from the domain.

Bridge prediction (30 sec, no wrong answers) Before Launch: what do you think the SA/Volume ratio simplifies to for a cube with side x ?

Write a guess — we compare in Launch

Launch — Example 2 + Example 6 (back-to-back)

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DOK 2

12 min

teacher models Ex 2 + Ex 6

DOMAIN RULES (mark these — assessment reference)

1. Domain = all reals EXCEPT zeros of **every** denominator in every step.
2. Write the simplified form AND the restriction list together.
3. Canceled factors still exclude their zeros.

MODELING RULES (mark these too)

1. Identify the ratio in words first (what / what).
2. Translate each quantity to a rational expression, with units.
3. Simplify, restrict domain, then interpret the simplified ratio in context.

Explore — TPS: 6 items in order (35 min)

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DOK 2

35 min

student work

Abstract forms first, then modeling, then assessment rehearsal. Never cut #19.

Try It 6 · SA/V comparison (two prisms) Which container has the smaller surface area-to-volume ratio?

Practice #14 · Domain restriction reasoning Why are restrictions needed to show $\frac{-6x^2 + 21x}{3x}$ and $-2x + 7$ are equivalent?

Practice #22 · Simplify + identify domain Simplify $\frac{y^2 - 5y - 24}{y^2 + 3y}$. State the domain.

DOK 2

5 min

DOK-2 reveal + Topic 4 assessment preview

Sentence frame for #19 (pair presents, class signals) “The denominator $x^3 + 3x^2 - 10x$ factors as _____.

The zeros are _____, _____, _____, so $x \neq$ _____.

A cubic denominator may have up to ____ restrictions.”

Topic 4 8-Q Assessment — Q#2 is this exact content Practice #19 directly rehearses Topic 4 LEHS Q#2.

Assessment cue: cubic denominator $\rightarrow$ factor completely $\rightarrow$ list ALL zeros.

Do NOT stop at one restriction.

Exit Ticket — Pre-Assessment Confidence Check

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DOK 1-2

3 min

not graded

Complete on your exit ticket

1. A rational model needs factoring because _____.
2. My simplified ratio is _____, valid when $x \neq$ _____.
3. In context this means _____.
4. One biggest thing I learned today: _____.